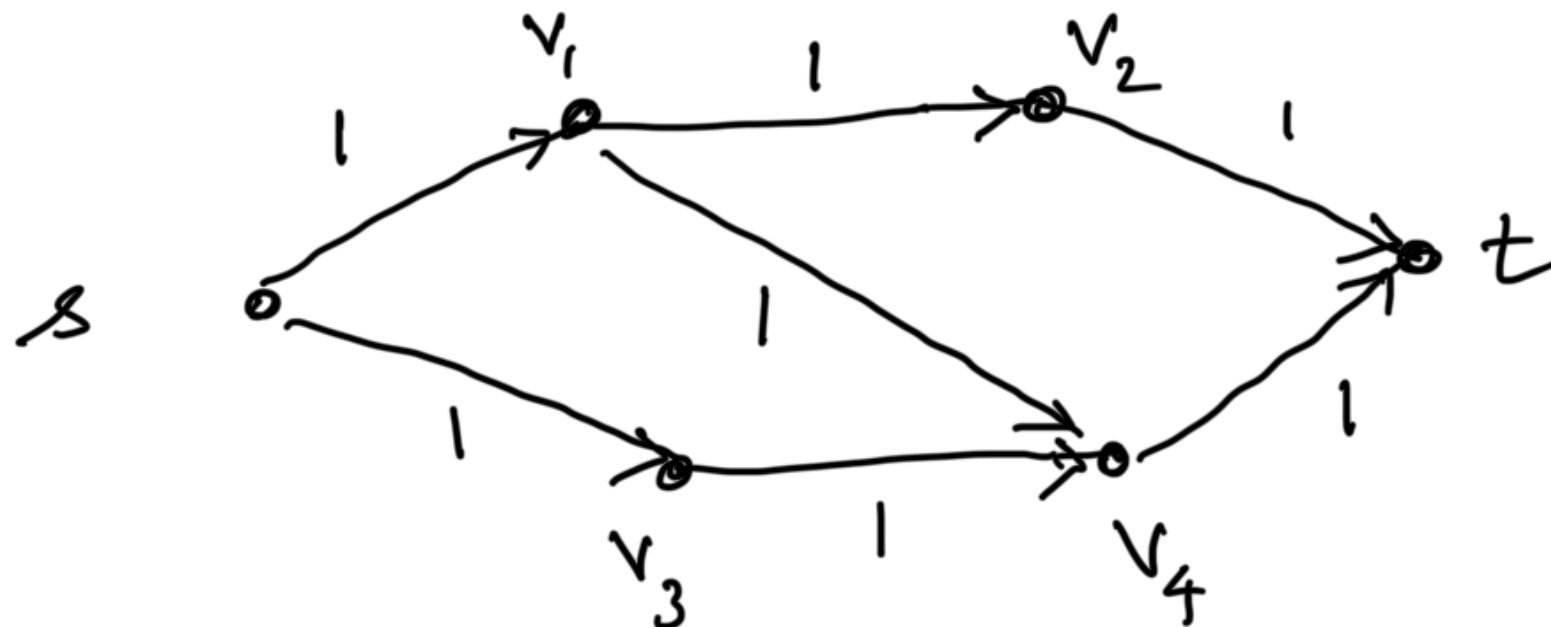


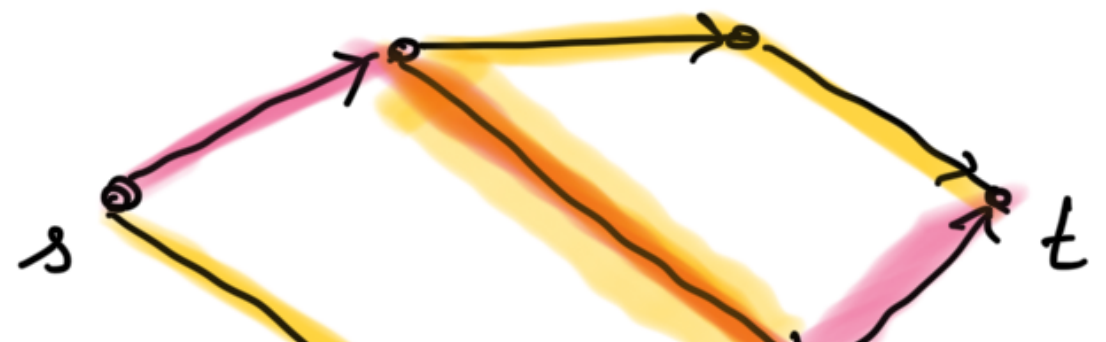
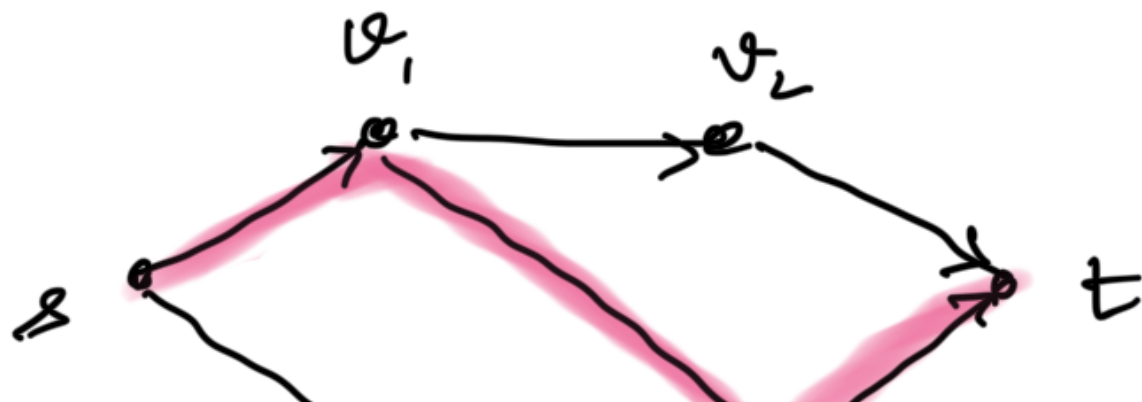
Lecture 25 DAA April 17

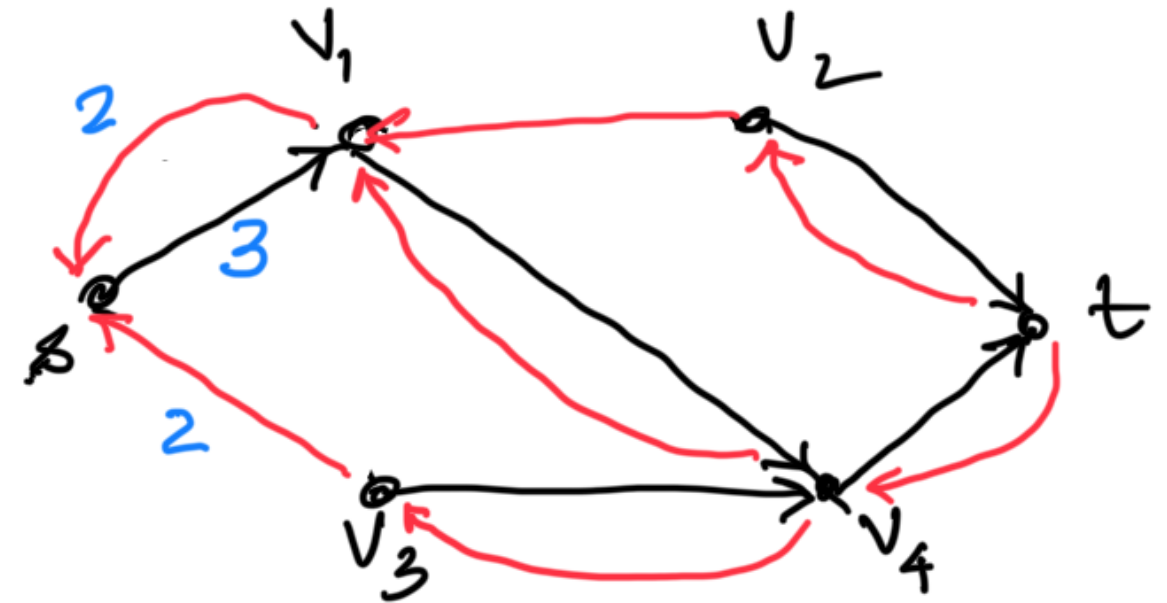
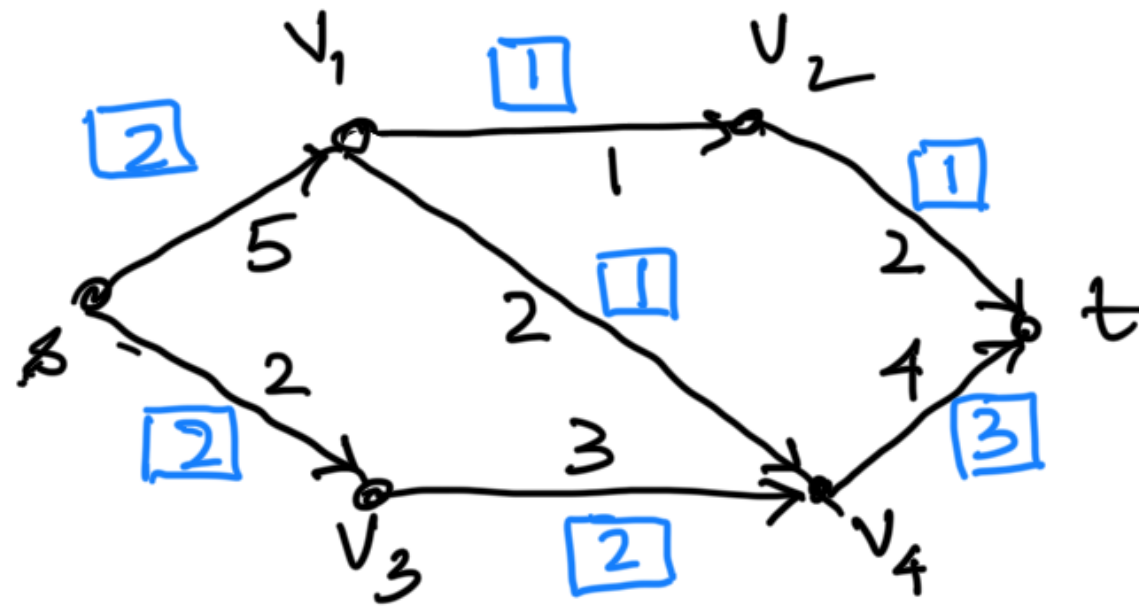


$$G = (V, E) \quad C: E \rightarrow \mathbb{R}^+ \quad f: E \rightarrow \mathbb{R}^+$$

capacity constraint: $\forall e \in E, \quad f(e) \leq c(e)$

conservation: $\forall v \in V - \{s, t\} \quad \sum_{e \in \text{In}(v)} f(e) = \sum_{e \in \text{Out}(v)} f(e)$



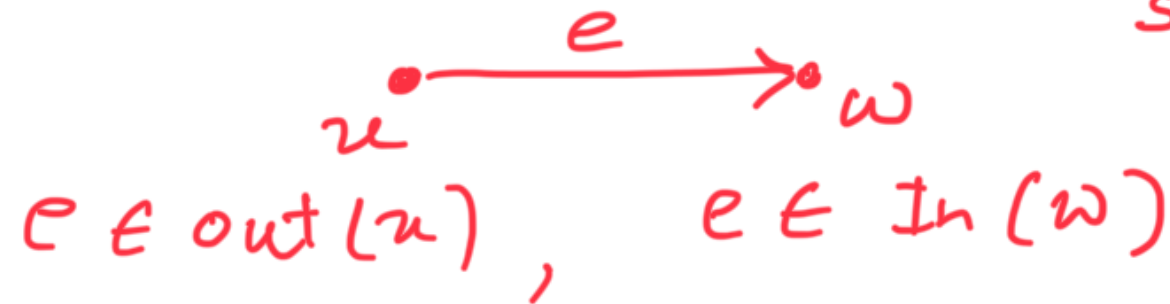


Augmenting path $s \rightsquigarrow t$ in Residual Graph $G(f)$
 flow $\nearrow$

$$\sum_{v \in V - \{s, t\}} \left(\sum_{e \in \text{out}(v)} f(e) - \sum_{e \in \text{in}(v)} f(e) \right) = 0 \quad \text{from conservation}$$

$$\Rightarrow \sum_{e \in \text{out}(s)} f(e) - \sum_{e \in \text{in}(s)} f(e) = \sum_{e \in \text{in}(t)} f(e) - \sum_{e \in \text{out}(t)} f(e)$$

Other $f(e)$ is counted once from $\text{in}(v)$ and $\text{out}(v)$
so - they cancel

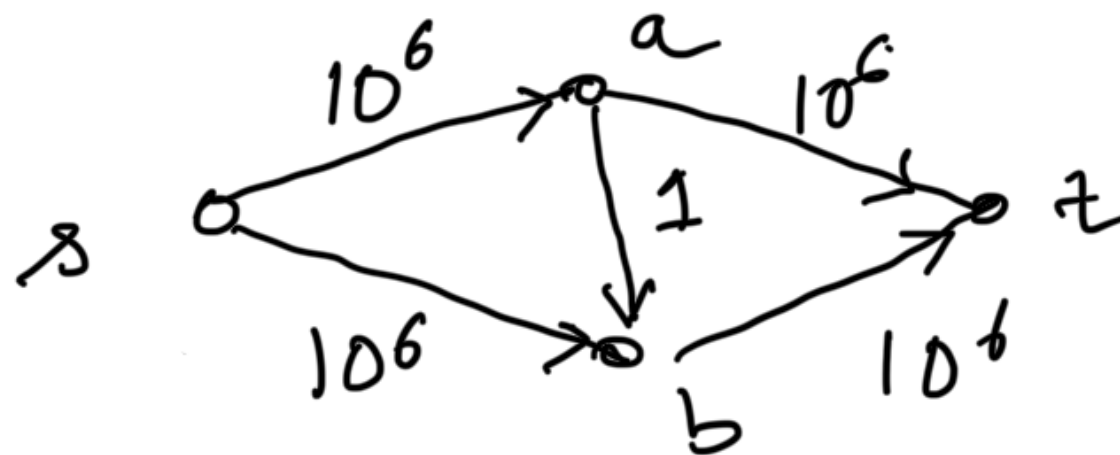


L.H.S. : net outflow from s R.H.S. : Net inflow to t

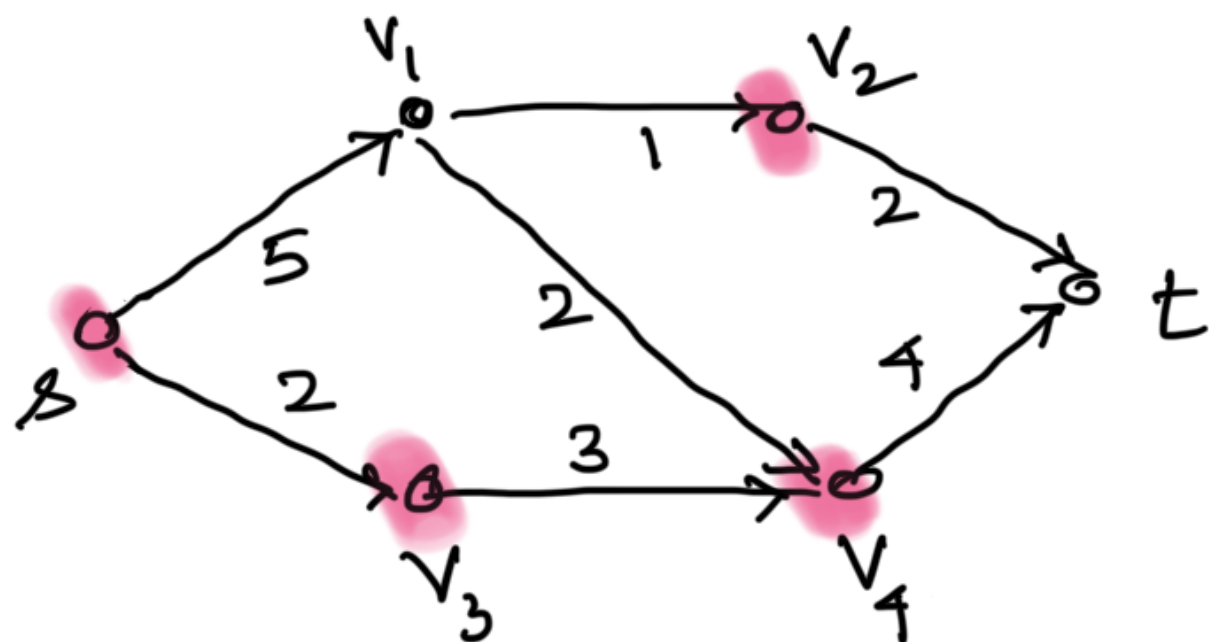
Flow problem : maximize Outflow s (Inflow t)

Observation : If there is an augmenting path,
the flow is not maximum.. Easy

$\Leftarrow$ No augmenting path $\Rightarrow$ Maximum flow?



How many iterations of augmentation?



Cut: $S, V-S$
 $s \in S, t \in V-S$

Cut edges: $\{uv \in E \mid u \in S, v \in V-S\}$
 $\in \text{Out}(S)$

Cut value = $\sum_{e \in \text{Cut edge}} c(e)$

$S = \{s, v_2, v_3, v_4\}$ $V-S = T = \{v_1, t\}$

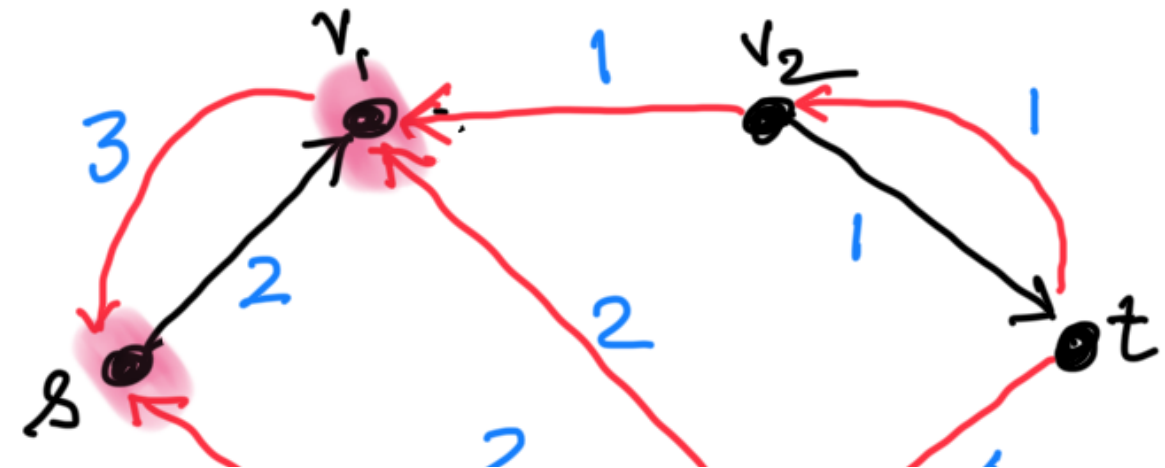
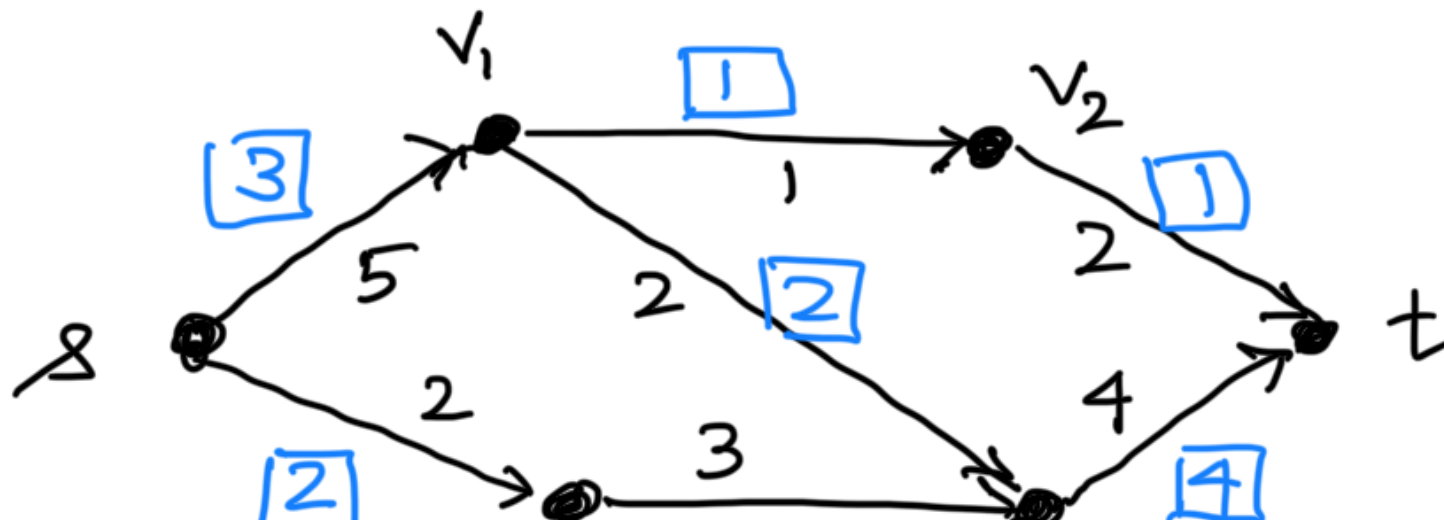
Cut edges = $\{(s, v_1), (v_2, t), (v_4, t)\}$ Cut value $5 + 2 + 4 = 11$

By adding - the flow of all edges incident on S , we get

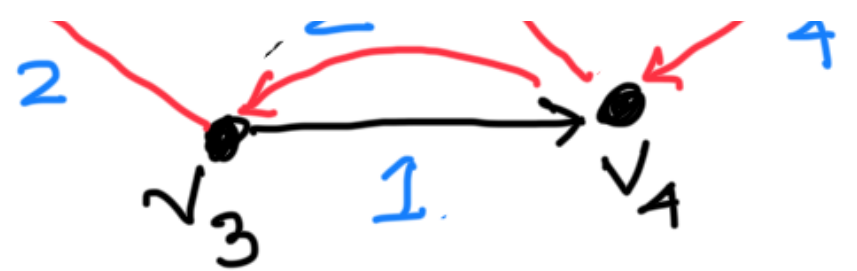
For any flow $F = \sum_{e \in \text{out}(S)} f(e) - \sum_{e \in \text{In}(S)} f(e) \rightarrow \{(v_1, v_2), (v_1, v_4)\}$

(Edges whose both endpoints are in S cancel)
 $\leq \sum_{e \in \text{out}(S)} f(e)$ since the second term > 0

$\Rightarrow \max \text{Flow} \leq \text{any Cut value (arbitrary cut)} \leq \min \{\text{All cut values}\}$
 ①

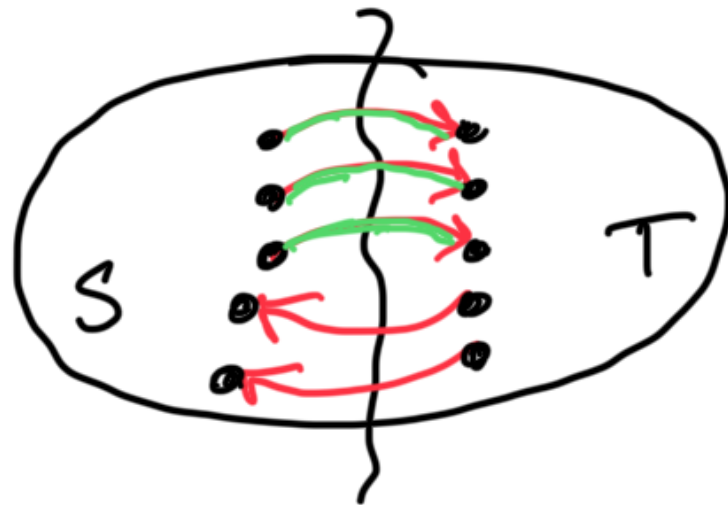


v_3 2 v_4



$S = \{v \mid \text{there is an augmenting path to } v\}$
 $= \{s, v_1\}$

Cut edges $\{(s, v_3), (v_1, v_2), (v_1, v_4)\}$



Claim: Forward edges are full capacity
 Back edges are 0

$$\Rightarrow \text{Flow } F^* = \sum_{e \in \text{out}(S)} f(e) = C^*$$

$$F^* \leq \max \text{Flow} \leq \min \text{cut} \leq \text{cut } C^*$$

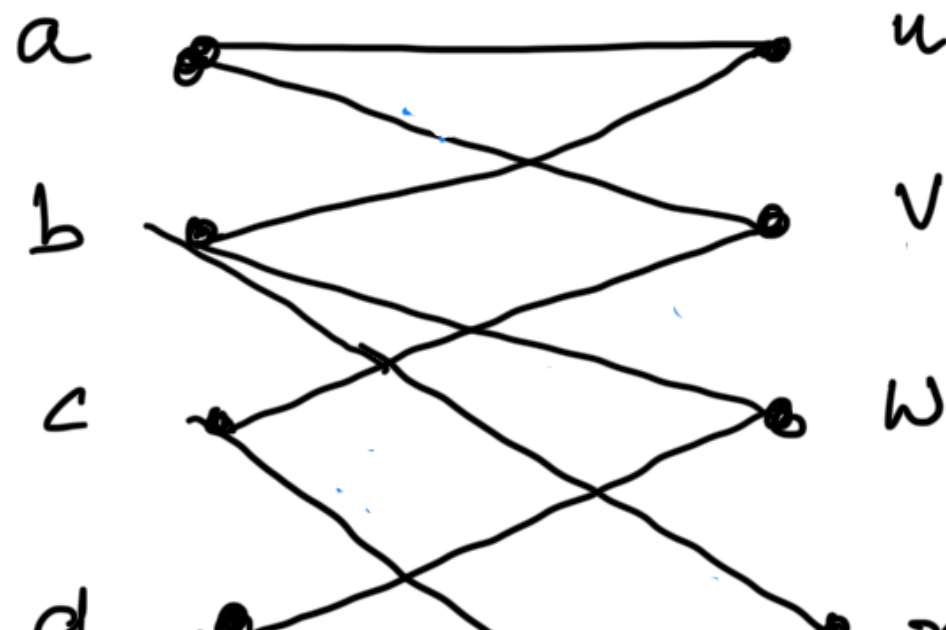
$$\Rightarrow \max \text{Flow} = \min \text{cut}$$

Ford-Fulkerson

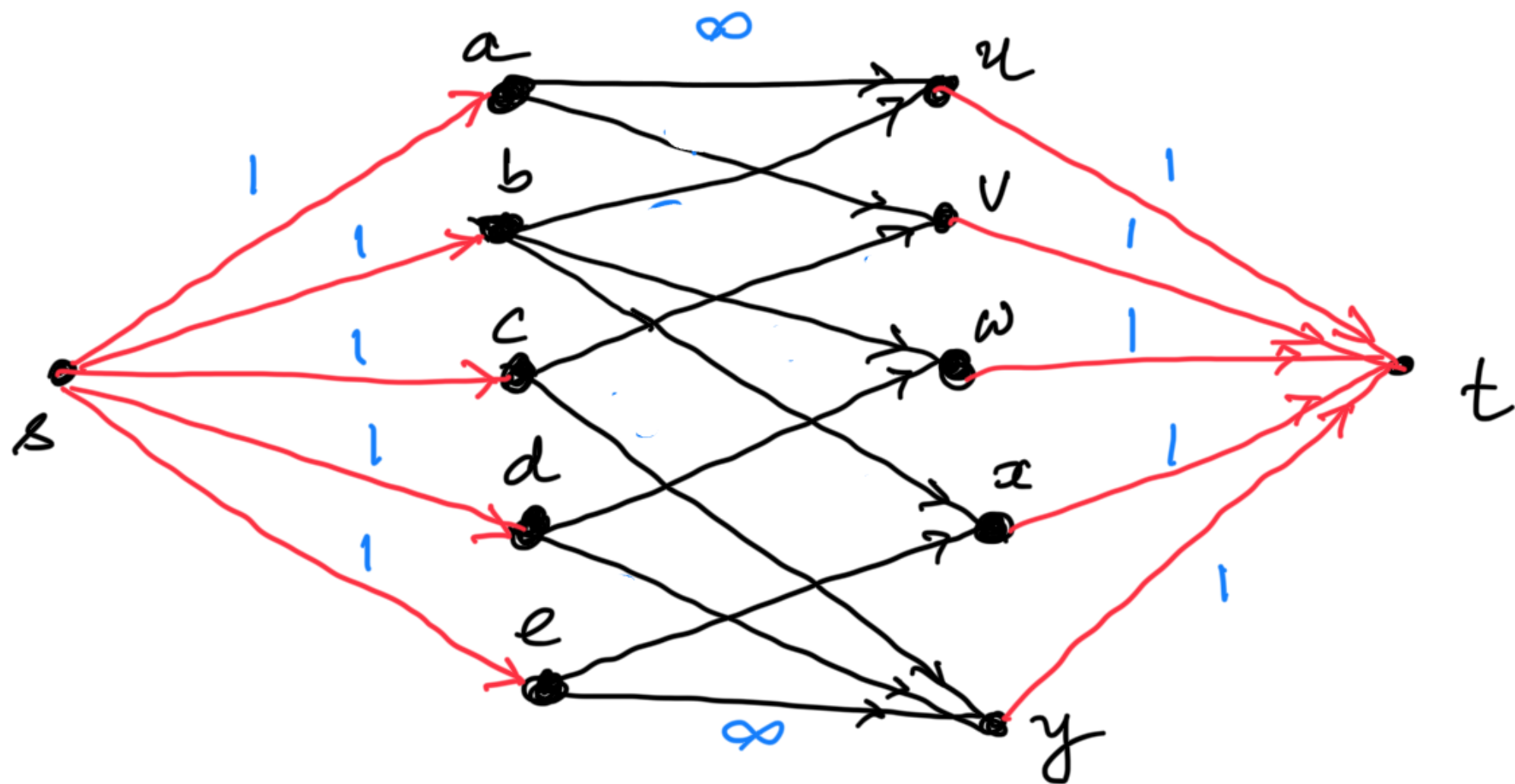
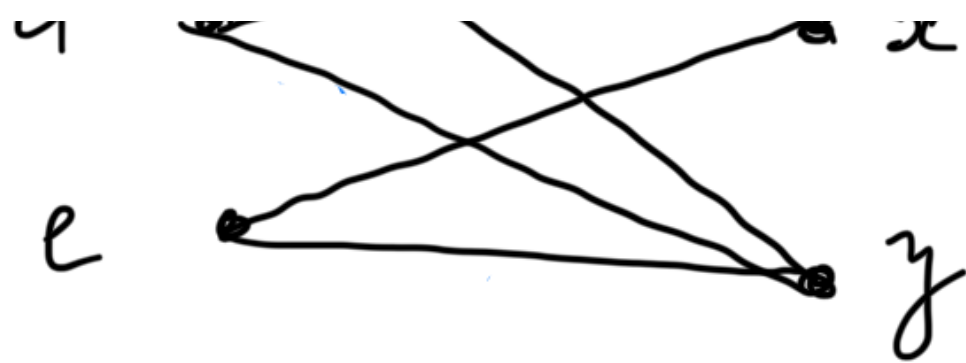
Not a guaranteed polynomial time

Edmond-Karp : Augmenting flow along shortest path results in a polynomial time algorithm

Applications of Max flow



Find the maxm
no. of vertex
disjoint edges



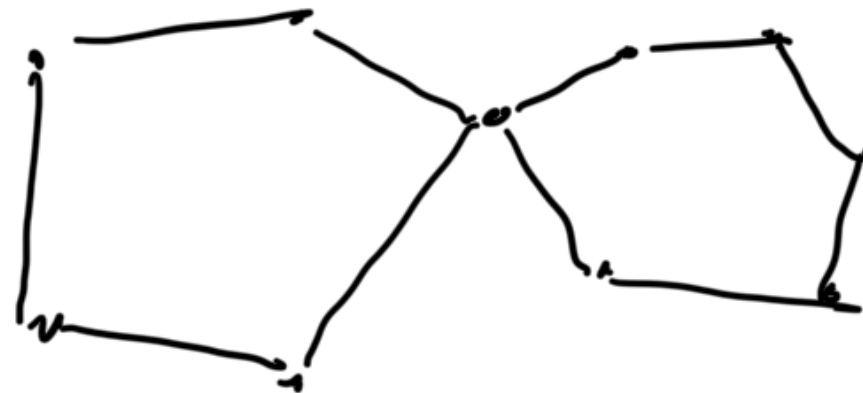
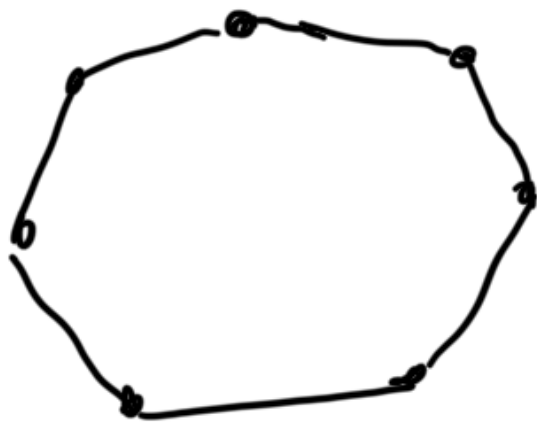
If there is a matching of size $k \rightarrow$ Handshaking

... is a matching of size $k \rightarrow$ flow of value k

If there is a flow of value $k \Rightarrow$ matching of size k
(All flow values integral)

k -vertex connectivity : At least k vertices must be removed to disconnect graph

k -edge connectivity : At least k edges must be removed to disconnect graph



Menger's theorem : The minimum no. of edges that must be removed to disconnect an s - t path

is equal to the # of edge disjoint paths between s and t .

Hall's theorem

Let G be a bipartite graph (L, R) , $L \cup R = V$.
There exists a perfect matching (all vertices are matched) iff for all $S \subseteq L$, $|\Gamma(S)| \geq |S|$
neighbours of S in R

Easy: If there is a perfect matching,
 $\Rightarrow |\Gamma(S)| \geq |S|$

Harder: If there is no perfect matching
 $\Rightarrow \exists S \subseteq L$ s.t. $|\Gamma(S)| < |S|$

König - Egervary theorem

Maximum matching = min Vertex Cover

